CENG381 Stochastic Processes

Poisson Process - Answer Key 1

Q1 Solution

a) Sales rate and expected time to 10th sale:

Thinning property: if arrivals are Poisson(lambda) and each is independently kept with probability p, the kept arrivals form a Poisson process with rate lambda' = lambda * p.

lambda' = 5 * 0.4 = 2 sales per hour

The time of the n-th sale, S_n = X_1 + ... + X_n, where each X_k ~ Exponential(lambda') are i.i.d. inter-sale times.

E[S_10] = E[X_1] + ... + E[X_10] = 10 * (1/lambda')
= 10 * (1/2) = 5 hours

b) P(no sales in first hour):

Using the Poisson PMF with rate lambda' = 2, time t = 1, n = 0:

P(N(1) = 0) = (lambda'*1)^0 * e^(-lambda'*1) / 0!
= 1 * e^(-2) / 1
= e^(-2)
~ 0.1353

There is about a 13.5% chance of making no sales in the first hour.

c) P(exactly 3 sales in first 2 hours):

Using the Poisson PMF with rate lambda' = 2, time t = 2, n = 3:

P(N(2) = 3) = (lambda'*2)^3 * e^(-lambda'*2) / 3!
= (2*2)^3 * e^(-4) / 6
= 4^3 * e^(-4) / 6
= 64 * e^(-4) / 6
e^(-4) ~ 0.01832
P(N(2) = 3) ~ 64 * 0.01832 / 6 ~ 1.1725 / 6 ~ 0.1954

There is about a 19.5% chance of exactly 3 sales in the first 2 hours.

Q2 Solution

a) Distribution of S_1 (time of first call):

$S_1 > t$ means no calls arrive in [0, t], i.e. $N(t) = 0$.

$$P(S_1 > t) = P(N(t) = 0) = \frac{(\lambda t)^0 e^{-\lambda t}}{0!} = e^{-\lambda t}$$

This is exactly the survival function of an Exponential(lambda) distribution: $P(\text{Exp}(\lambda) > t) = e^{-\lambda t}$.

Therefore $S_1 \sim$ Exponential(lambda = 4 calls/hour). (OK)

b) P(first call within 15 minutes = 0.25 hours):

$$P(S_1 \le 0.25) = 1 - P(S_1 > 0.25) = 1 - e^{-\lambda \cdot 0.25} = 1 - e^{-4 \cdot 0.25} = 1 - e^{-1}$$

$$\sim 1 - 0.3679 = 0.6321$$

There is about a 63.2% chance of the first call arriving within 15 minutes.

c) Waiting time from S_1 to S_2:

Key property used: INDEPENDENT INCREMENTS.

The number of calls in any interval depends only on the LENGTH of the interval, not on when it starts or what happened before.

In particular, the time from S_1 to S_2 (the 2nd inter-arrival time X_2) is independent of S_1 and has the SAME distribution:

$$X_2 = S_2 - S_1 \sim \text{Exponential}(\lambda = 4)$$

This is the MEMORYLESS property of the exponential distribution.

$$E[S_2] = E[S_1] + E[X_2] = 1/\lambda + 1/\lambda = 2/\lambda = 2/4 = 0.5 \text{ hours} = 30 \text{ minutes}$$

Q3 Solution

a) Binomial model X_m ~ Binomial(m, lambda*T/m):

In each of the m sub-intervals, there is either 0 or 1 arrival,
each with probability p_m = lambda*T/m. The total count X_m is
the sum of m independent Bernoulli(p_m) trials:

X_m ~ Binomial(m, lambda*T/m)

E[X_m] = m * p_m = m * (lambda*T/m) = lambda*T

Var[X_m] = m * p_m * (1 - p_m)
= lambda*T * (1 - lambda*T/m)
-> lambda*T as m -> infinity

b) Convergence to Poisson(lambda*T) as m -> infinity:

Binomial PMF for a fixed n:

P(X_m = n) = C(m,n) * (lambda*T/m)^n * (1 - lambda*T/m)^(m-n)

Expand each factor as m -> infinity:

C(m,n) = m*(m-1)*...*(m-n+1) / n!
~ m^n / n! (for fixed n, large m)

(lambda*T/m)^n = (lambda*T)^n / m^n

(1 - lambda*T/m)^(m-n) ~ (1 - lambda*T/m)^m
-> e^(-lambda*T) as m -> infinity
[using the standard limit: (1 - x/m)^m -> e^(-x)]

Combining:

P(X_m = n) -> (m^n / n!) * ((lambda*T)^n / m^n) * e^(-lambda*T)
= (lambda*T)^n * e^(-lambda*T) / n!

This is exactly the Poisson(lambda*T) PMF. (OK)

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c) Forward equations and verification:

Derivation of dp_j(t)/dt:

In the small interval [t, t+Delta]:

P(N(t+Delta)=j) = P(N(t)=j) * P(0 arrivals in Delta)
    + P(N(t)=j-1) * P(1 arrival in Delta)
    + P(N(t)<=j-2) * P(2+ arrivals in Delta)

Using P(0 arrivals in Delta) = 1 - lambda*Delta + o(Delta)
and P(1 arrival in Delta) = lambda*Delta + o(Delta):

p_j(t+Delta) = p_j(t)*(1-lambda*Delta) + p_{j-1}(t)*(lambda*Delta) + o(Delta)

Rearrange and divide by Delta, then take Delta -> 0:

dp_j(t)/dt = -lambda*p_j(t) + lambda*p_{j-1}(t)   for j >= 1

Verification that p_j(t) = (lambda*t)^j * e^(-lambda*t) / j! satisfies this:

Left side: dp_j/dt
  = d/dt [(lambda*t)^j * e^(-lambda*t) / j!]
  = [j*lambda*(lambda*t)^(j-1)*e^(-lambda*t)
    - lambda*(lambda*t)^j * e^(-lambda*t)] / j!
  = lambda*e^(-lambda*t) / j! * [j*(lambda*t)^(j-1) - (lambda*t)^j]

Right side: -lambda*p_j + lambda*p_{j-1}
  = -lambda*(lambda*t)^j*e^(-lambda*t)/j!
   + lambda*(lambda*t)^(j-1)*e^(-lambda*t)/(j-1)!
  = lambda*e^(-lambda*t) * [-(lambda*t)^j/j! + (lambda*t)^(j-1)/(j-1)!]
  = lambda*e^(-lambda*t)/j! * [-(lambda*t)^j + j*(lambda*t)^(j-1)]

Left side = Right side (OK)

The Poisson PMF satisfies the forward equations.